

AssureX Claim Engine

User Guide

Group	NN_DevStorm
Batch	2609E2'
Faculty	Sir Minhaj
Repository	https://github.com/Saba1512006/assurex-claim-engine
Live application	https://assurex.pythonanywhere.com
Date	29 September 2026

Student ID	Name	Role
Student1525913	Saba Noor	Team lead · integration, frontend direction, QA, documentation
Student1524867	Muhammad Sami	Backend and ML pipeline developer · deployment
Student1509222	Muhammad Ghanyan	Frontend design (Figma, colour palette, mobile checks)
Student1505677	Muhammad Sami ur Rehman	Testing (sample claims, Chrome and mobile)

Abstract

AssureX Claim Engine evaluates warranty claims for consumer electronics, home appliances and industrial tools. A customer or service center registers a product from its receipt (read by OCR), files a claim with evidence, and the application pre-processes the claim into model features. A calibrated Python classification model (HistGradientBoosting, 89.3% test accuracy on 225 unseen claims) scores the structured record; the same claim is drawn as a Claim Summary Card and classified by a Google Teachable Machine model (93.8% test accuracy). The two predictions and their confidences are compared (Strong Match, Acceptable Match, Weak Match, Model Disagreement, Uncertain Result), the category's warranty policy is executed from JSON configuration, contradictions, missing documents and duplicates are detected, and a decision table produces Likely Valid, Likely Invalid or Manual Review Required. Anything uncertain goes to a human reviewer, and every step is recorded in an audit trail. The application has four roles (customer, service-center staff, claim reviewer, administrator), 295 automated tests and 34 browser tests.

How to use the application (SRS §1.10 item 10), from registration to exporting a claim report, with screens of the working application for every role, and how to test the deployed version.

Contents

1. Execution instructions and user guide
2. Deployed application

1. Execution instructions and user guide

Quick start

Requirements: Python 3.11 or 3.12 (3.11 recommended), Git. Works on Windows, macOS and Linux.

```
git clone https://github.com/Saba1512006/assurex-claim-engine.git
cd assurex-claim-engine
python -m venv venv
source venv/bin/activate           # Windows: .\venv\Scripts\Activate.ps1
pip install -r requirements.txt
python database/seed.py           # creates the database + demo data, writes .env with a
random SECRET_KEY
python src/app.py                 # http://127.0.0.1:5000
```

Run the tests: `python -m pytest -q` (295 tests, plus 34 browser tests; latest results in `reports/test_results.txt`). Browser suite (13 journeys with accessibility checks): `pip install -r requirements-dev.txt && python -m playwright install chromium && python -m pytest -m e2e tests/e2e`. Production: `gunicorn wsgi:app` (Render uses `render.yaml`; PythonAnywhere's WSGI file imports `application` from `wsgi.py`). Set `SECRET_KEY` in the environment — the app refuses to start without one.

How to use it

Task	Where
Register / sign in	/register , /login
Register a product and its warranty	<i>Products > Register product</i> . Upload the receipt first — it is read by OCR and fills the form; check every value. Choose standard or extended cover.
Add warranty information / documents	Product page > <i>Product documents</i> (receipt, warranty card, serial photo) — used by every claim on that product
Create a claim	<i>New claim</i> — 4 steps: product → what happened → evidence → review
Upload documents	Wizard step 3, or the claim page > <i>Evidence</i> while the claim is a draft, in review, or waiting for information
Verify extracted information	Wizard step 3 shows the values read from the receipt in editable fields; on the claim page each document has <i>Extracted details</i> . Corrections are audited.
Submit	Wizard <i>Submit claim</i> (or <i>Save as draft</i> then <i>Submit claim</i> on the claim page)
Python prediction & confidence	Claim page > verdict, left column: predicted class and probability for all three classes (calibrated: 0.80 means right about 80% of the time), plus the inputs that moved it most
Claim Summary Card	Claim page > <i>What the image model saw</i> , with the tiles that drove its answer outlined
Teachable Machine prediction	Claim page > verdict, right column (or “Model unavailable” until the export is installed)
Compare both models	The agreement gauge between the two columns: both needles, the gap Δ , the minimum-confidence mark and the consistency status
Warranty-rule results	Claim page > <i>Why this decision</i> : failed and review rules first, passed rules underneath
Contradictions / duplicates	Same list; duplicates also flag the claim in lists
Manual-review queue	Reviewer or admin > <i>Review queue</i> , then the workbench (/reviewer/claim/<id>)
Administrator dashboard	Admin > <i>Overview</i> (filters by category, status and date) and <i>Analytics</i>
Try other thresholds safely	Admin > <i>What-if simulator</i>

Task	Where
Evaluate many records at once	Admin > <i>Batch evaluation</i> (CSV in the training-split format)
Choose what customers see	Admin > <i>Policies</i> > <i>What customers see</i> (model probabilities on or off)
Track a claim	Claim page > <i>Track progress</i> , or the 8-stage bar at the top of the claim page
Export a claim report	Claim page > <i>Download report (PDF)</i> ; bulk CSV or Excel (.xlsx) from Admin > <i>Overview, Analytics, Audit</i> and claim <i>Search</i>

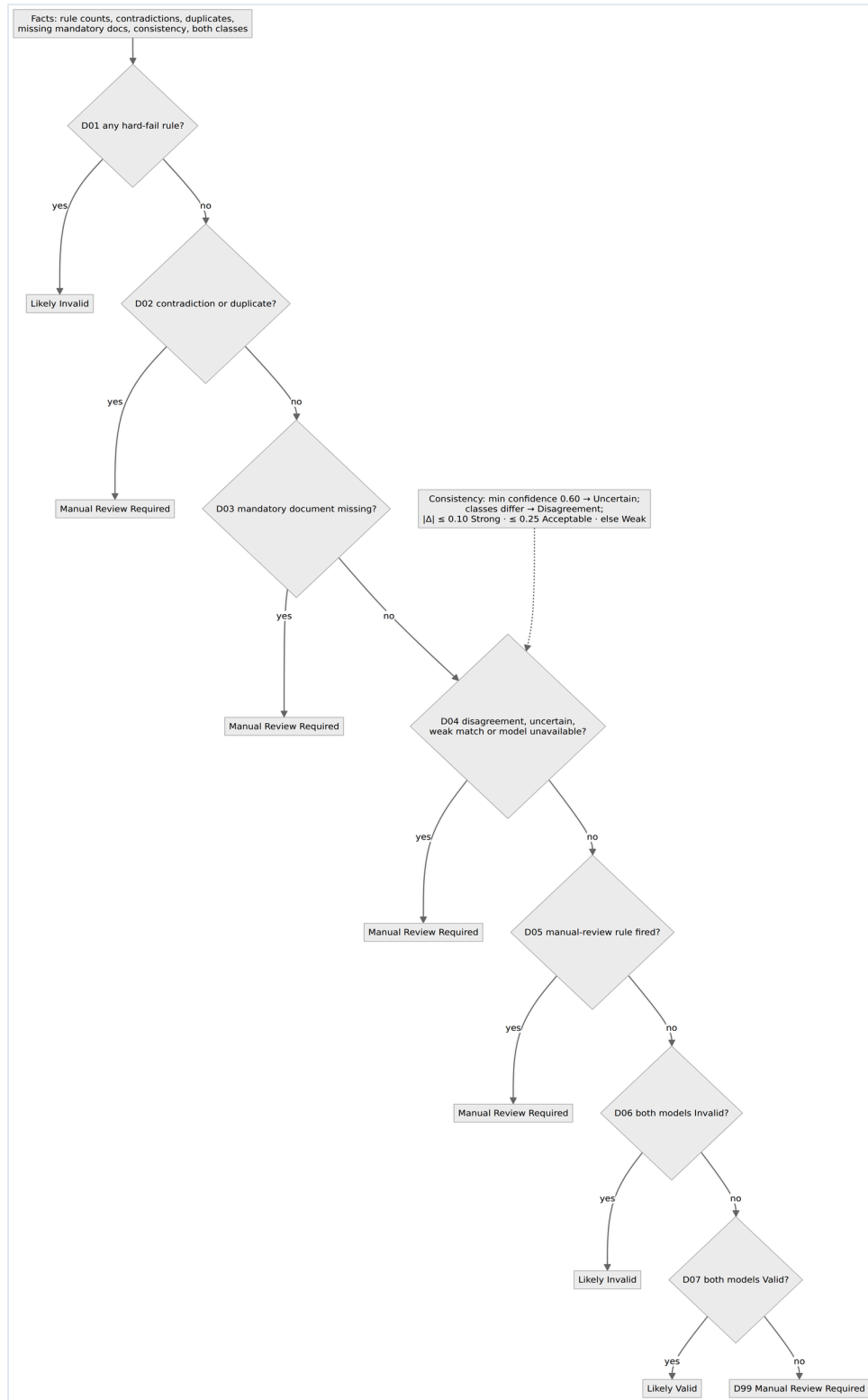
The 11 SRS demonstration cases

`python database/seed.py` creates one real claim for each case, submitted through the live pipeline: valid, invalid (excluded damage), manual review (unknown cause), expired warranty, missing receipt, duplicate (reused receipt and invoice), contradictory dates, serial mismatch, unauthorised repair, boundary date (inside the grace period) and a model disagreement (*KitchenPro Oven 45L*: the image model says Valid, the Python model Invalid, so the claim goes to a reviewer). The same cases exist as feature records in `sample_claims/` and are pinned by `tests/test_demonstration_cases.py`.

How a claim is decided

1. **Evidence** — files are checked by magic bytes (not the extension), stored under random names, SHA-256 fingerprinted, and OCR-read (PDF text via pdfplumber; images via Tesseract when installed).
2. **Contradictions** — claim date vs purchase date, fault date vs purchase/claim dates, repair dates, and the registered serial / model / invoice / purchase date vs every value read from the evidence.
3. **Duplicates** — the same file hash on another product or claim, an invoice number registered to another product, another open claim on the same serial, a near-identical fault description, the same claimant repeating the same fault within 30 days.
4. **Features** — `src/core/features.py` builds exactly the 20 columns the model was trained on (product age, days to expiry, reporting delay, missing-document count, flags ...).
5. **Rules** — `policies/<category>.json` decides each rule's severity: *hard fail*, *manual review*, *warning* (advisory) or off. Coverage, grace period, reporting deadline, excluded causes, evidence rules and repair history are all in the file.
6. **Python model** → class + 3 probabilities. **Claim Summary Card** (600×600, facts only, never a prediction) → **Teachable Machine** → class + 3 probabilities.
7. **Consistency** (`config/decision_policy.json`): top confidence below 0.60 → *Uncertain Result*; different classes → *Model Disagreement*; $|\Delta| \leq 0.10$ *Strong*, ≤ 0.25 *Acceptable*, otherwise *Weak Match*.
8. **Decision table** — first matching row wins: D01 hard fail → Likely Invalid; D02 contradiction or duplicate, D03 missing mandatory document, D04 disagreement / uncertain / weak / model unavailable, D05 manual-review rule → Manual Review Required; D06 both Invalid → Likely Invalid; D07 both Valid → Likely Valid; otherwise D99 Manual Review Required. `routing` maps each recommendation to a status.

Every evaluation is a new, immutable database row holding the exact inputs, both model versions (file hashes), the card hash, the thresholds and the decision trace — updating a model never changes a recorded result.



Decision flow

Security

Deny-by-default access control ([config/rbac.json](#)): permission → record scope (*own, service center, assigned or queue, all*) → constraints (reviewers can't decide claims they filed, a claim can only move along allowed transitions, overrides need a 20-character reason, admins can't demote themselves or the last admin). Out-of-scope records return 404. Users are reloaded from the database on every request and a role change or disable signs them out everywhere. Also: generic login errors, 5-attempt lockout, session rotation, CSRF on every form, rate-limited sign-in, strict Content-Security-Policy (no

inline or third-party scripts), magic-byte upload validation, CSV formula-injection protection and an append-only audit trail. Details: [documentation/RBAC_DESIGN.md](#). Landing hero background: [documentation/HERO_ENGINE.md](#).

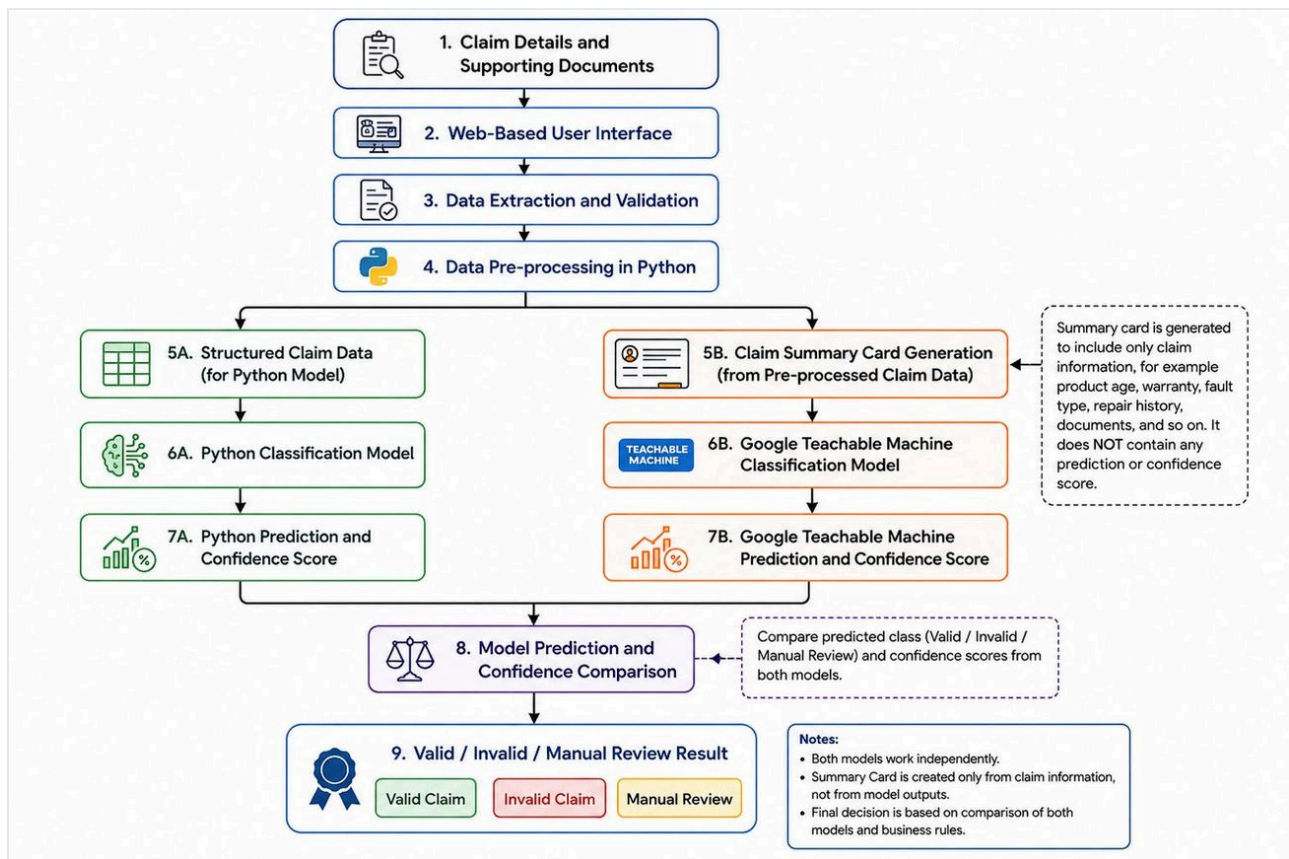
Repository layout

config/	rbac.json, decision_policy.json, config.py		
policies/	one warranty policy per category (JSON)		
data/	raw + split CSVs, summary_cards/{train,val,test}, mapping file, statistics, data dictionary		
dataset_generator/	generator_v2.py (records), build_cards_v2.py (cards)		
notebooks/	train_python_v2.py, evaluate_gtm.py		
model/	python_model/ (pipeline + model card), teachable_machine/ (export goes here)		
src/core/	vocab, features, python + TM runtimes, card renderer, consistency, decision table, pipeline		
src/rules/	policy store, rule engine, contradiction + duplicate detectors, validation		
src/security/	RBAC policy engine and Flask guards		
src/services/	claim lifecycle, documents, OCR glue, alerts, analytics, exports, PDF report, explanations		
src/api/	Flask blueprints	templates/, static/	UI
database/	seed.py	tests/	pytest suites (tests/e2e:
browser journeys)			
scripts/	subset_icons.py (icon-font subset)	docs/screenshots/	images produced by the e2e suite
reports/	comparison report, project report builder		
documentation/	project report, blog, evidence, installation, test cases, diagrams		

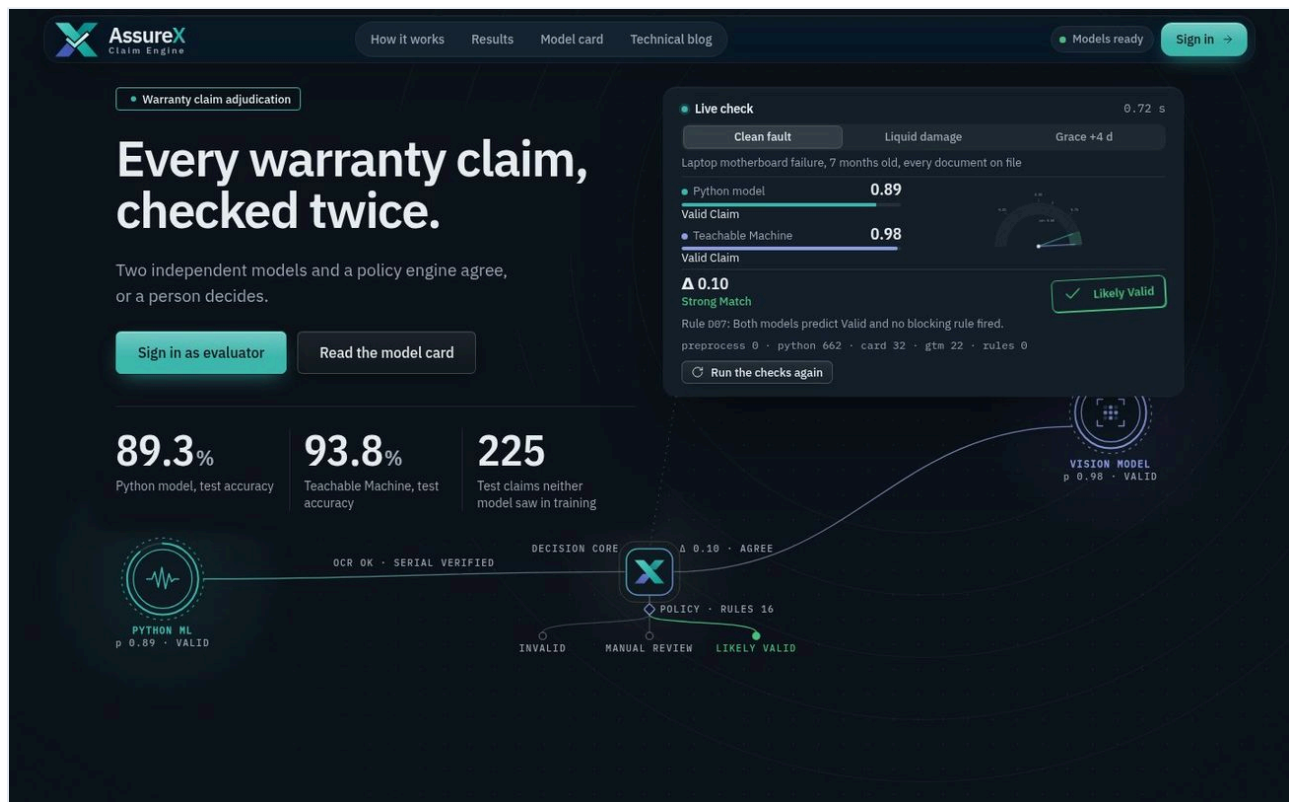
Known limitations

- The Teachable Machine model is trained in the browser and must be retrained by hand whenever the card design or the policy limits shown on the card change.
- The dataset is synthetic (generated from the policy with 4% label noise). Real claims will need retraining and re-calibration.
- OCR of **image** receipts needs the Tesseract binary; without it, images are accepted and the user types the details. PDF receipts with a text layer are read without Tesseract.
- SQLite is the default database; use [DATABASE_URL](#) (e.g. PostgreSQL) and a shared rate-limit store ([RATELIMIT_STORAGE_URI](#)) for multi-worker production.
- The saved Python model needs scikit-learn 1.9.1, which needs Python 3.11 or newer. The PythonAnywhere free account used for the live demo runs Python 3.10 (scikit-learn $\leq$ 1.7.2) with a 512 MB disk quota, so both models can show as *Unavailable* there and claims then go to manual review, as designed. The full pipeline runs locally with the pinned requirements (Python 3.11, see [documentation/INSTALLATION.md](#)).

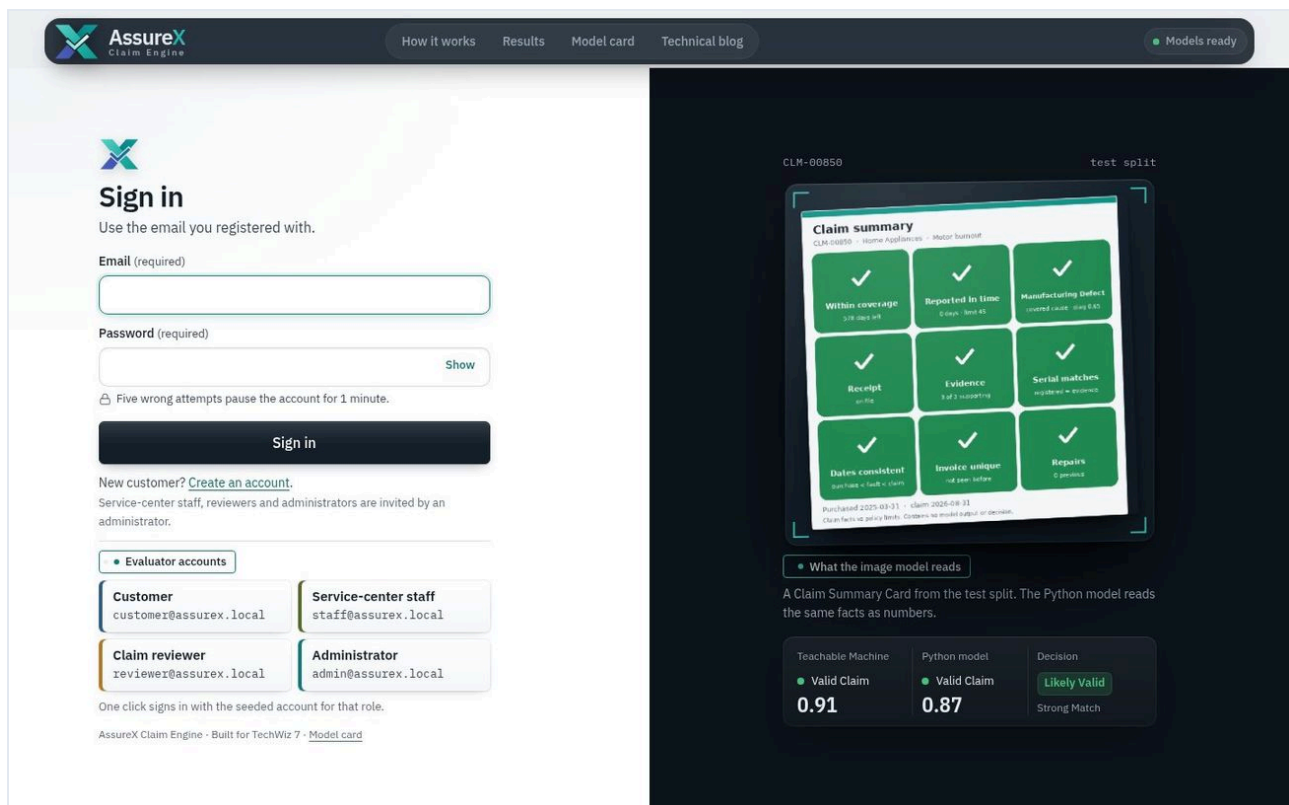
Screens of the working application



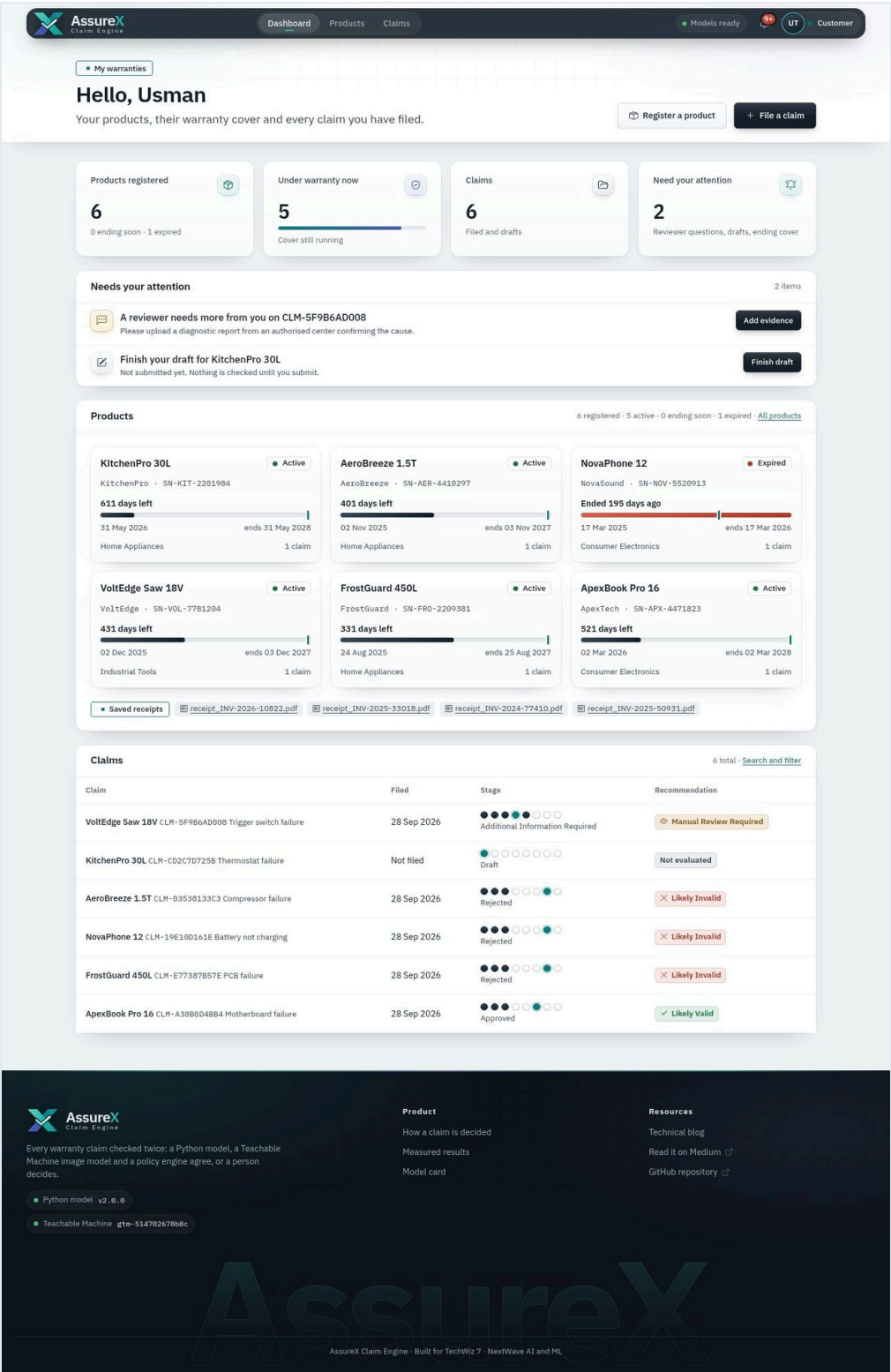
00_system_architecture.jpg — System architecture



01_landing.jpg — Landing page: headline, measured results, the live check deciding a real sample claim, and the dual-engine background (Python ML and vision model feeding the decision core, policy engine and outcomes)



02_login.jpg — Sign-in beside a scanned test-split Claim Summary Card, with one-click evaluator accounts



03_customer_dashboard.jpg — Customer dashboard: at-a-glance KPIs, pending actions, products and claims

Overview

Claims

Review queue

Analytics

Policies

Access

Audit

Models ready

AK

Administrator

Claims / CLM-26CC70D39B

NovaPods Max

Speaker malfunction · filed for Hira Siddiqui

Manual Review

Medium risk

Track progress

Download report (PDF)

CLM-26CC70D39B NovaPods Max Consumer Electronics

Decided in 0.25 s

Python model

v2.0.0

Manual Review 0.94

Valid 0.03

Invalid 0.04

Manual Review 0.94

Teachable Machine

gtm-514702678b8c

Manual Review 0.55

Valid 0.11

Invalid 0.34

Manual Review 0.55

0.50

0.25

0.75

0.60

0.39

Uncertain Result

Lowest top-class confidence 0.55 is below the 0.60 minimum.

Manual review required

Why this decision

The models disagree, are not confident, or one is unavailable. Rule D04 matched (D01-D03 did not).

D01

D02

D03

D04

Claim filed inside the post-expiry grace period. Claim filed 4 days after expiry, inside the 7-day grace period.

15 warranty checks passed

Evidence still needed

None.

What moved the Python model

Each input replaced by its typical training value; bar = change in the probability of "Manual Review".

Days to warranty end: -4 (typical 221)

+0.88

Purchase price: 329 (typical 619.22)

+0.14

Product category: Consumer Electronics (typical Industrial Tools)

-0.04

What the image model saw

Claim summary

CLM-26CC70D39B · Consumer Electronics · Speaker malfunction

Grace period

Reported in time

Manufacturing Defect

Receipt on file

Evidence

Serial matches

Dates consistent

Invoice unique

Repairs

The card holds claim facts only — no prediction, confidence or decision. Teal outlines mark the tiles that most drove the image model's answer.

Serial matches (registered = evidence)

-0.26

Grace period (4 of 7 grace days)

-0.02

Pipeline timings (ms): preprocess 5 · python 124 · card 27 · gtm 91 · rules 0 · total 250

Evidence

Each file is fingerprinted (SHA-256); receipts are read by OCR

Receipt

Warranty card

Damage photo

Serial photo

Repair report missing

Warranty card

warranty_card.png · 10.3 KB · 4bc65a02771a · 28 Sep 2026

Open

Extracted details — OCR not available for this file

Damage / fault photo

damage_photo.png · 10.7 KB · 6ba8fc72e06f · 28 Sep 2026

Open

Serial-number photo

serial_photo.png · 12.4 KB · 3a4c4e45e07b · 28 Sep 2026

Open

Reviewer decision

Unassigned

Assign reviewer

Claim details

Product

NovaPods Max

PRD-0BEDBF91BC

Category

Consumer Electronics

Serial

SN-NOV-8830154

05_claim_manual_review_d04.jpg — Claim page, *Manual Review Required* (rule D04): Δ 0.39, *Uncertain Result* because the image model's top confidence is below 0.60; the decision trace, the Claim Summary Card the image model saw and what moved each model (top of the page)

Dashboard
Products
Claims

Models ready
99
UT
Customer

Claims / CLM-E77387857E

FrostGuard 450L

PCB failure · filed for Usman Tariq by Sana Malik

Rejected
High risk

Track progress
Download report (PDF)

CLM-E77387857E
FrostGuard 450L
Home Appliances
Decided in 0.24 s

Python model
v2.0.0
Invalid Claim 0.86
Valid 0.04
Invalid 0.86
Manual Review 0.10

Teachable Machine
gtm-514702678b8c
Invalid Claim 0.99
Valid 0.00
Invalid 0.99
Manual Review 0.00

0.50
0.25
0.75
min 0.60

Δ 0.13
Acceptable Match
Both predict Invalid Claim; confidence difference 0.13.
Likely invalid

Why this decision

A policy hard-fail rule was triggered. Rule D01 matched.

D01

✗ Damage cause is excluded by the policy and confirmed by diagnosis. 'Water Ingress' is excluded for Home Appliances and the diagnosis confirms it ($0.81 \geq 0.55$).

► 15 warranty checks passed

Evidence still needed
None.

What moved the Python model
Each input replaced by its typical training value; bar = change in the probability of "Invalid Claim".

Damage cause: **Water Ingress** (typical Manufacturing Defect) +0.81

Diagnostic confidence: **0.81** (typical 0.58) +0.07

Purchase price: **1250** (typical 619.22) -0.06

What the image model saw

Claim summary

CLM-E77387857E · Home Appliances · PCB failure

Within coverage 30 days aft	Reported in time 4 days limit 6s	Water Ingress excluded cause · diag 0.81
Receipt on file	Evidence 3 of 3 supporting	Serial matches registered = evidence
Dates consistent prior date < fault < claim	Invoice unique not seen before	Repairs 0 previous

Purchased 2025-08-24 · claim 2026-09-28
Claim facts vs policy limits. Contains no model output or decision.

The card holds claim facts only — no prediction, confidence or decision. Teal outlines mark the tiles that most drove the image model's answer.

Water Ingress (excluded cause · diag 0.81) -0.99

Dates consistent (purchase < fault < claim) -0.96

Invoice unique (not seen before) -0.91

Pipeline timings (ms): preprocess 5 · python 125 · card 26 · gtm 83 · rules 0 · total 242

Evidence

Each file is fingerprinted (SHA-256); receipts are read by OCR

Receipt
Warranty card
Damage photo
Serial photo
Repair report missing

Warranty card
warranty_card.png · 10.0 KB · 932a6001ffe0 · 28 Sep 2026
Open

Extracted details — OCR not available for this file

Damage / fault photo
damage_photo.png · 10.4 KB · 624e02016c1b · 28 Sep 2026
Open

Claim details

Product	FrostGuard 450L PRD-1F735B2469
Category	Home Appliances
Serial	SN-FRO-2209381
Purchased	24 Aug 2025 · National Appliance Depot
Warranty	Active until 25 Aug 2027
Fault started	22 Sep 2026

06_claim_likely_invalid_d01.jpg — Claim page, *Likely Invalid* (rule D01): rejected by a hard-fail warranty rule (top of the page)

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Dashboard
Products
Claims

Models ready
9+
UT
Customer

CLM-A3880D48B4 > Tracking

Where is my claim?

ApexBook Pro 16 · Motherboard failure

Open the full claim

Draft
28 Sep 15:10

Submitted
28 Sep 15:10

Under Evaluation
28 Sep 15:10

More information needed

Manual Review

Approved
28 Sep 15:10

Rejected

Closed

Right now

Approved

Automated recommendation

Likely Valid

Approved. The service center will contact you about the repair or replacement.

History

Approved

Automated recommendation Likely Valid (rule D07: Both models predict Valid and no blocking rule fired)

28 Sep 2026 15:10 · System

Under Evaluation

Automated evaluation started

28 Sep 2026 15:10 · System

Submitted

Claim submitted

28 Sep 2026 15:10 · Service-center staff

Draft

Claim created

28 Sep 2026 15:10 · Service-center staff

Every warranty claim checked twice: a Python model, a Teachable Machine image model and a policy engine agree, or a person decides.

Python model v2.0.0

Teachable Machine gtm-514702678b8c

Product

How a claim is decided
Measured results
Model card

Resources

Technical blog
Read it on Medium
GitHub repository

AssureX

AssureX Claim Engine · Built for TechWiz 7 · NextWave AI and ML

07_claim_tracking.jpg — Eight-stage claim tracking

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Dashboard
Products
Claims

Models ready
9+
UT
Customer

Products and warranties

Every registered product with how much of its warranty is left.

+ Register a product

Search

Category

All categories

Warranty

All
Active
Expiring soon
Expired

Search

6 products.

KitchenPro 30L
Active

KitchenPro · SN-KIT-2201984

611 days left

31 May 2026 ends 31 May 2028

Home Appliances 1 claim

AeroBreeze 1.5T
Active

AeroBreeze · SN-AER-4410297

401 days left

02 Nov 2025 ends 03 Nov 2027

Home Appliances 1 claim

NovaPhone 12
Expired

NovaSound · SN-NOV-5520913

Ended 195 days ago

17 Mar 2025 ends 17 Mar 2026

Consumer Electronics 1 claim

VoltEdge Saw 18V
Active

VoltEdge · SN-VOL-7781204

431 days left

02 Dec 2025 ends 03 Dec 2027

Industrial Tools 1 claim

FrostGuard 450L
Active

FrostGuard · SN-FRO-2209381

331 days left

24 Aug 2025 ends 25 Aug 2027

Home Appliances 1 claim

ApexBook Pro 16
Active

ApexTech · SN-APX-4471823

521 days left

02 Mar 2026 ends 02 Mar 2028

Consumer Electronics 1 claim

Every warranty claim checked twice: a Python model, a Teachable Machine image model and a policy engine agree, or a person decides.

Python model v2.0.0
Teachable Machine gtm-514702678b8c

Product

How a claim is decided
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08_products.jpg — Registered products with warranty status

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Dashboard
Products
Claims

Models ready
98
UT
Customer

Products > PRD-42220D1221

ApexBook Pro 16

Active
Consumer Electronics
ApexTech - ABP-16
SN-APX-4471823

File a claim

Warranty

ApexTech manufacturer warranty

521

02 Mar 2028

24 months

days of cover left

cover ends

standard term

Started 02 Mar 2026

Ends 02 Mar 2028 - then 7 days grace

Service center

Metro Authorised Service Center

Covered faults

Screen flickering, Motherboard failure, Battery not charging, Speaker malfunction, Touch panel unresponsive, Wi-Fi/Bluetooth failure

Not covered

Liquid ingress or immersion; Screen shattering from drops; Power surge / overvoltage damage; Cosmetic casing wear; Commercial rental use

Report faults within

30 days of noticing them

Repairs

Repairs must be carried out by an authorised service center using genuine parts.

Purchase

Product ID	PRD-42220D1221
Bought	02 Mar 2026
Retailer	City Electronics Mall
Price	1,899.00
Invoice	INV-2025-48213
Registered	28 Sep 2026

Claims

1 on this product

Claim	Stage	Recommendation	Filed
Motherboard failure CLM-A3880D48B4	Approved	Likely Valid	28 Sep 2026

Product documents

Every claim on this product uses them

Purchase receipt / invoice
receipt_INV-2025-48213.pdf · 38e492957611 · 28 Sep 2026 · text read

Open
Replace
Remove

Document type

Purchase receipt / invoice

File

Choose File | No file chosen

PDF, JPG or PNG, at most 10 MB.

Upload document

Repair history

No repairs on record

Every warranty claim checked twice: a Python model, a Teachable Machine image model and a policy engine agree, or a person decides.

Python model v2.0.0
Teachable Machine gtm-514702678b8c

Product

How a claim is decided
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09_product_warranty.jpg — Product and warranty page: coverage, documents and repair history

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Claims

Review queue

Models ready

BA

Claim reviewer

Review queue

Claims the automated checks could not settle: low confidence, model disagreement, missing evidence, duplicates, contradictions or a policy review rule.

Unassigned 5

Assigned to me 0

Waiting on the customer 1

Decided

Category

Risk

Model agreement

All categories

Any risk

Any

5 claims

Highest risk first, then oldest

Claim	Why it is here	Python	Teachable Machine	Agreement	Waiting
ApexBook Pro 16 CLM-BC91073EBB Kamran Ali High risk	D02 Contradiction or duplicate indicator found ; possible duplicate	Manual Review 0.85	Manual Review 0.96	Acceptable Match	0 d Take
VividTab 11 CLM-FD180BC3BC Hira Siddiqui High risk	D02 Contradiction or duplicate indicator found 1 contradiction	Manual Review 0.55	Manual Review 0.85	Uncertain Result	0 d Take
CleanCycle 8kg CLM-15580B2CE4 Hira Siddiqui Medium risk	D03 A mandatory document is missing Supporting evidence; missing receipt	Manual Review 0.96	Manual Review 0.98	Strong Match	0 d Take
NovaPods Max CLM-26CC70D39B Hira Siddiqui Medium risk	D04 The models disagree, are not confident, or one is unavailable Grace-period claim	Manual Review 0.94	Manual Review 0.55	Uncertain Result	0 d Take
KitchenPro Oven 45L CLM-FE40F6C0B8 Kamran Ali Medium risk	D04 The models disagree, are not confident, or one is unavailable	Invalid Claim 0.91	Valid Claim 0.97	Model Disagreement	0 d Take



Every warranty claim checked twice: a Python model, a Teachable Machine image model and a policy engine agree, or a person decides.

Python model v2.0.0

Teachable Machine gtm-514702678b8c

Product

How a claim is decided

Measured results

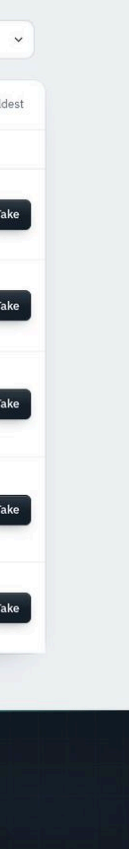
Model card

Resources

Technical blog

Read it on Medium

GitHub repository



AssureX Claim Engine - Built for TechWiz 7 - NextWave AI and ML

AssureX

Claim Engine

Claims

Review queue

Models ready

BA

Claim reviewer

Review queue

CLM-26CC70D39B

NovaPods Max • Speaker malfunction

Manual Review

Medium risk

Unassigned

Filed 28 Sep 2026 • waiting 0 d

Shortcuts

Claim page

Next claim

Evidence

4 files

Warranty card

NovaSound NovaPods Max

Synthetic demo image

Warranty card

warranty_card.png • 4bc65a92771a

Open

Damage / fault photo

Open

CLM-26CC70D39B NovaPods Max

Consumer Electronics

Decided in 0.25 s

0.50

0.25

0.75

0.60

0.39

Uncertain Result

Lowest top-class confidence 0.55 is below the 0.60 minimum.

Python model

Teachable Machine

v2.0.0

gtn-514702678b8c

Manual Review 0.94

Manual Review 0.55

Warranty

Ended 4 days before today (inside the 7-day grace period)

24 Sep 2025

ends 24 Sep 2026 + 7 d grace

Fault started

22 Sep 2026

Reported

28 Sep 2026 (6 d later; limit 30)

Cause given

Manufacturing Defect

Diagnosis

0.75 • Service-center technician

Customer's account

Customer

Hira Siddiqui

Other claims

2

Approve

Reject

Request information

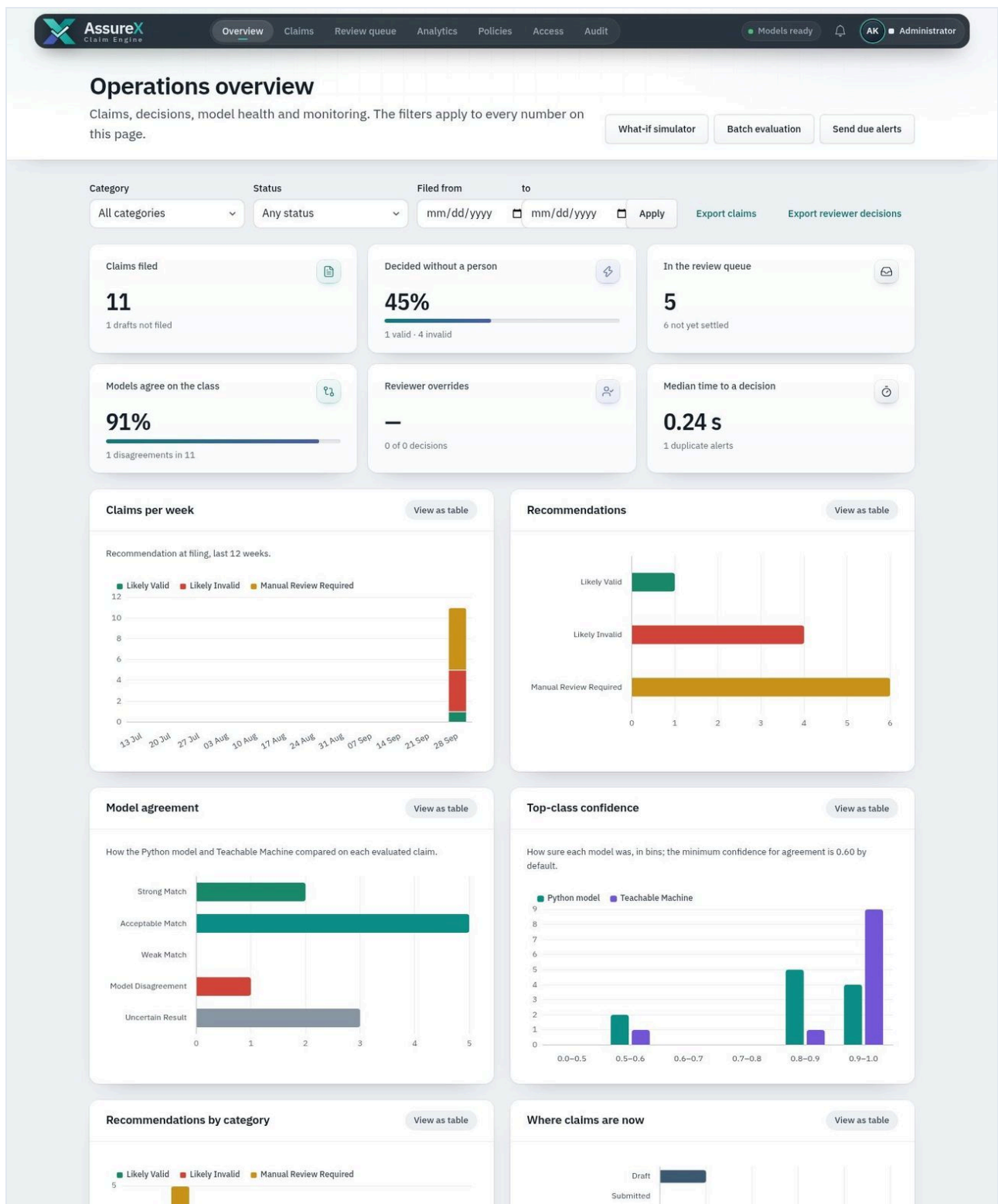
Comment (shown to the customer, kept in the audit trail)

Recommendation: Manual Review Required

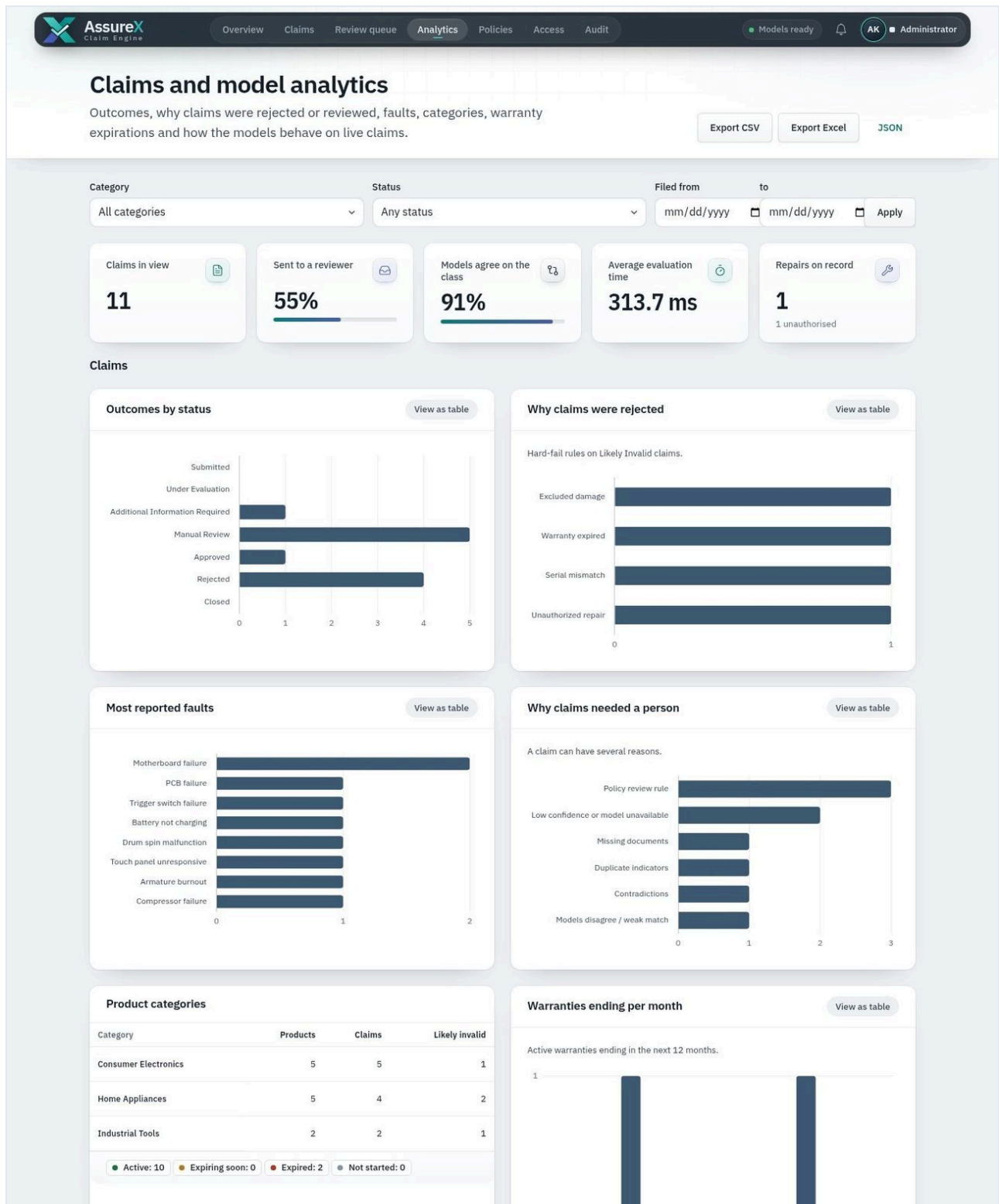
Record decision

Take this claim

11_reviewer_decision.jpg — Reviewer workbench: evidence, the verdict with both models' scores, warranty facts, and the decision bar with keyboard shortcuts



12_admin_overview.jpg — Operations overview: filters, KPI cards, claims per week, recommendations, model agreement and confidence charts (top of the page)



13_analytics_model_evidence.jpg — Analytics and model evidence (top of the page)

[Overview](#)
[Claims](#)
[Review queue](#)
[Analytics](#)
[Policies](#)
[Access](#)
[Audit](#)

Models ready

AK

Administrator

Warranty policies and decision settings

Changes apply to new evaluations immediately; recorded results never change. Every save is versioned in the audit trail.

Try thresholds in the what-if simulator

Consumer Electronics
Home Appliances
Industrial Tools

Standard Consumer Electronics Warranty Policy

policies/consumer_electronics.json

Standard coverage (months)

12

Grace period after expiry (days)

on the card

7

Report faults within (days)

on the card

30

Repeat repairs before review

on the card

2

Diagnosis needed to confirm an exclusion

on the card

0.55

An excluded cause with a diagnosis at or above this is a hard fail; below it the claim goes to a reviewer.

Excluded causes

on the card

☐ Manufacturing Defect
☐ Normal Wear and Tear
☒ Electrical Surge
☐ Mechanical Stress
☒ Accidental Drop
☒ Water Ingress
☐ Unknown / Not Sure

Rules and severity

Hard fail: Likely Invalid · Manual review: a reviewer · Warning: shown, never blocks

Rule	Severity
Claim filed after the warranty and its grace period ended WARRANTY_EXPIRED	Hard fail
Claim filed inside the post-expiry grace period GRACE_PERIOD	Manual review
Warranty ends within 14 days of the claim WARRANTY_ENDING_SOON	Warning
Product was repaired by an unauthorised service center UNAUTHORIZED_REPAIR	Hard fail
Damage cause is excluded by the policy and confirmed by diagnosis EXCLUDED_DAMAGE	Hard fail
Damage cause is excluded but the diagnosis is not conclusive EXCLUDED_DAMAGE_UNCONFIRMED	Manual review
Serial number differs from the receipt and serial photo SERIAL_MISMATCH	Hard fail
Serial number mismatch that the evidence cannot confirm SERIAL_UNVERIFIED	Manual review
Fault reported after the policy's reporting deadline LATE_REPORTING	Hard fail
Fault reported in the last week before the deadline NEAR_REPORTING_DEADLINE	Warning
Fault is not in the policy's list of covered faults UNCOVERED_FAULT	Manual review
Cause unknown and diagnostic confidence below 0.50 UNKNOWN_CAUSE	Manual review
Product has reached the repeat-repair review threshold REPEAT_REPAIRS	Manual review
Product was already replaced once under warranty PRIOR_REPLACEMENT	Manual review
Two or more supporting documents are missing INCOMPLETE_EVIDENCE	Manual review
One supporting document is missing SUPPORTING_DOCUMENT_MISSING	Warning

Changes

None yet

Edit a value to see the change here before saving.

Save Consumer Electronics policy

Discard

Version history

Consumer Electronics

No saved changes yet: this is the policy as shipped.

Fixed by the policy file

Covered faults

Screen flickering,
Motherboard failure,
Battery not charging,
Speaker malfunction,
Touch panel unresponsive, Wi-Fi/Bluetooth failure

Mandatory

receipt

Supporting

warranty card, serial photo, damage photo

14_policy_editor.jpg — Warranty policy and rule-severity editor, decision settings, version history (top of the page)

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[Analytics](#)
[Policies](#)
[Access](#)
[Audit](#)

Models ready

AK Administrator

Overview > Models

Models

Each model is versioned by the hash of its file, and every stored prediction records the version that made it. Installing a new model never changes past results.

Public model card

Python classification model

Running

Version	v2.0.0+514e2fb68286
Algorithm	Hist gradient boosting, sigmoid-calibrated (5-fold)
Trained with	scikit-learn 1.9.1 - seed 42
Inputs	20 features: 8 numeric, 10 yes/no, 2 categories
Test accuracy	89.3% (target: 85%)
Macro F1 · ROC-AUC	89.2% · 0.9475
Calibration (ECE)	0.0818

Retrain with python notebooks/train_python_v2.py.

Google Teachable Machine

Running

Version	gtm=514702678b8c
File	model_unquant.tflite · floating point · input 224x224
Test accuracy	93.8% macro F1 93.8%
Validation accuracy	97.3%
Agrees with Python	92.9% of test claims

Re-run the evaluation on validation and test cards

Train and install the Teachable Machine model

- Download the training cards: 2100 images in three class folders, two variations per training claim. Validation and test cards are left out.

Download training cards (.zip)
- Open teachablemachine.withgoogle.com, then Image Project, Standard image model.
- Name the classes **Valid Claim**, **Invalid Claim** and **Manual Review** and upload each folder into its class.
- Train. 50 epochs, batch 16 and learning rate 0.001 are a good start.
- Export Model, TensorFlow Lite, Floating point, Download, then unzip.
- Upload model_unquant.tflite and labels.txt here and run the evaluation.

Model file (.tflite)

Choose File

No file chosen

labels.txt

Choose File

No file chosen

Checked before it replaces the current model: TensorFlow Lite header, three labels matching the classes, and a test run. The previous model is archived.

Install the model

Sample Claim Summary Cards

600 × 600 · claim facts only · one renderer for training and live claims

Claim summary

CLM-00004_v1.jpg

Valid Claim

Claim summary

CLM-00006_v1.jpg

Invalid Claim

Claim summary

CLM-00003_v1.jpg

Manual Review

Every warranty claim checked twice: a Python model, a Teachable Machine image model and a policy engine agree, or a person decides.

Python model v2.0.0

Teachable Machine gtm=514702678b8c

Product

How a claim is decided
Measured results
Model card

Resources

Technical blog
Read it on Medium
GitHub repository

AssureX

AssureX Claim Engine · Built for TechWiz 7 · NextWave AI and ML

15_models_teachable_machine.jpg — Model status, Teachable Machine installer and evaluation, sample cards

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AssureX

Claim Engine

Overview

Claims

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Models ready

AK

Administrator

Access control

Who can sign in, what each role may do and which records it reaches. A role change signs the person out everywhere so the new limits apply at once.

Customer

3

People with this role

Service-center staff

1

People with this role

Claim reviewer

2

People with this role

Administrator

1

People with this role

People

7 shown

Name or email

Role

Status

Filter

Person

Role

Status

Areeba Khan (you)

admin@assurex.local USR - DA76B1D522

Administrator

Active

Sign out

Bilal Ahmed

reviewer@assurex.local USR - 9FDF3E72BE

Claim reviewer

Active

Sign out

Disable

Nadia Qureshi

reviewer2@assurex.local USR - 5666027353

Claim reviewer

Active

Sign out

Disable

Hira Siddiqui

hira@example.com USR - CDF1DFA271

Customer

Active

Sign out

Disable

Kamran Ali

kamran@example.com USR - 47755FAED6

Customer

Active

Sign out

Disable

Usman Tariq

customer@assurex.local USR - C433C0AD90

Customer

Active

Sign out

Disable

Sana Malik

staff@assurex.local USR - 78D2F53CB8 · Metro Au

Service-center staff

Active

Sign out

Disable

Invite someone

Full name (required)

Work email (required)

Role

Customer

They get a one-time password shown once to you, and must set their own at first sign-in. Public sign-up only creates customer accounts.

Create the account

Recent blocked attempts

No blocked attempts yet.

Service centers

Harbor Tech Repairs

Karachi · +92 21 3555 0200 · SC-D5A3C34C92

Metro Authorised Service Center

Lahore · +92 42 3555 0100 · SC-61C50C31D1

Center name

City

Phone

Add the center

What each role can do

Scope = which records a permission reaches · config/rbac.json

Permission

Customer

Service-center staff

Claim reviewer

Administrator

admin.dashboard

—

—

—

all

alert.dispatch

—

—

—

all

analytics.read

—

—

—

all

audit.read

—

—

—

all

claim.create

own

service center

—

—

claim.read

own

service center

assigned or queue

all

claim.submit

own

service center

—

—

claim.update_draft

own

service center

—

—

document.correct_ocr

own

service center

—

—

document.delete

own

—

—

—

document.read

own

service center

assigned or queue

all

16_access_control.jpg — Access control: people, roles, invitations, service centers, blocked attempts (top of the page)

Audit trail

Every sign-in, upload, OCR correction, prediction, status change, reviewer decision, configuration change and blocked attempt. Nothing here can be edited or deleted.

Export CSV

Export Excel

Action

Role

Actor

Record or details

From

to

All actions

Any role

USR- or email

CLM-, a file hash, a rule

mm/dd/yyyy

mm/dd/yyyy

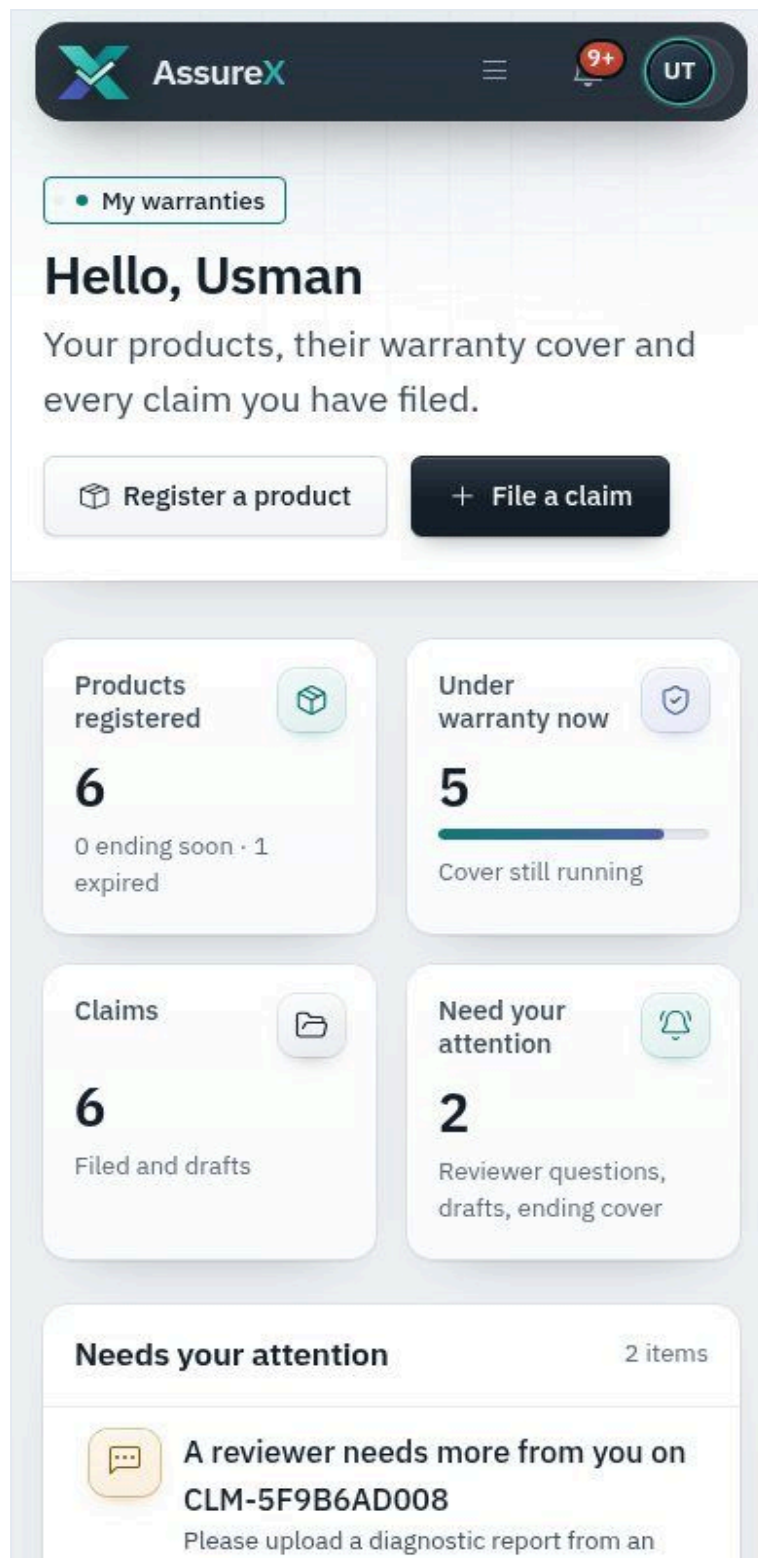
Search

72 events

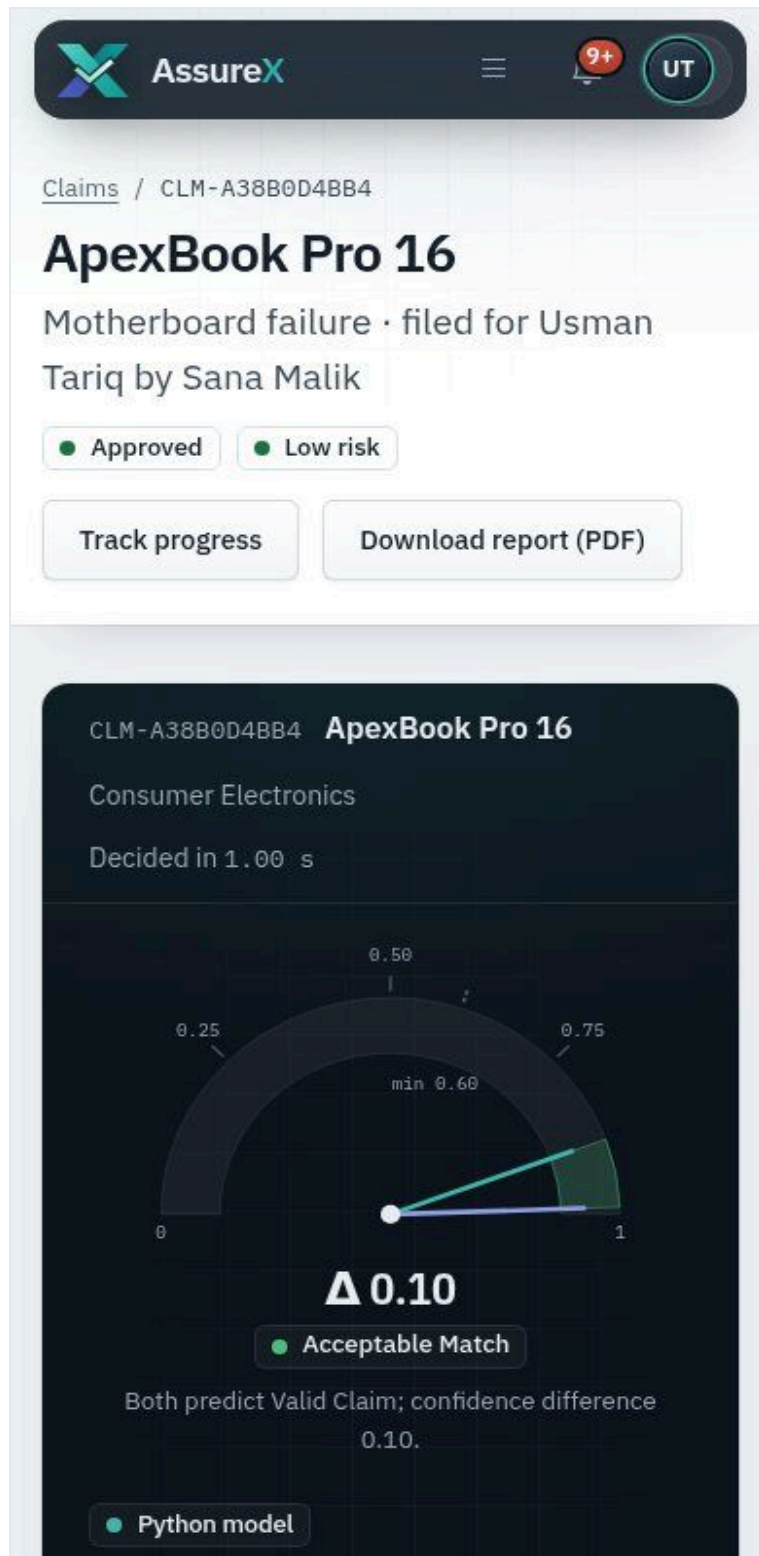
Newest first · times in UTC

Time	Action	Actor	Record	Details	IP	
28 Sep 15:11:55	LOGIN_SUCCESS	Bilal Ahmed Claim reviewer	User USR-9FDF3E72BE		127.0.0.1	
28 Sep 15:11:35	LOGIN_SUCCESS	Areeba Khan Administrator	User USR-DA76B1D522		127.0.0.1	
28 Sep 15:11:26	LOGIN_SUCCESS	Usman Tariq Customer	User USR-C433C0AD90		127.0.0.1	
28 Sep 15:10:08	REVIEWER_DECISION	Bilal Ahmed Claim reviewer	Claim CLM-5F9B6AD008	{ "from_status": "Manual Review", "to_status": "Additional..." }	127.0.0.1	
28 Sep 15:10:08	STATUS_CHANGE	Bilal Ahmed	Claim CLM-5F9B6AD008	{ "from": "Manual Review", "to": "Additional Information Required", "..."		
28 Sep 15:10:08	REVIEWER_ASSIGNED	Bilal Ahmed Claim reviewer	Claim CLM-5F9B6AD008	{ "reviewer": "USR-9FDF3E72BE", "previous": null }		
28 Sep 15:10:08	CLAIM_SUBMITTED	Kamran Ali Customer	Claim CLM-FE40F6C0B8	{ "first_submission": true }		
28 Sep 15:10:08	STATUS_CHANGE	System	Claim CLM-FE40F6C0B8	{ "from": "Under Evaluation", "to": "Manual Review", "note": "..."		
28 Sep 15:10:08	FINAL_DECISION	Kamran Ali Customer	Claim CLM-FE40F6C0B8	{ "decision": "Manual Review Required", "rule": "D04", "..."		
28 Sep 15:10:08	MODEL_PREDICTION	Kamran Ali Customer	Claim CLM-FE40F6C0B8	{ "python": { "predicted_class": "Invalid Claim", "top_confidence": "..."		
28 Sep 15:10:08	STATUS_CHANGE	System	Claim CLM-FE40F6C0B8	{ "from": "Submitted", "to": "Under Evaluation", "note": "Automated..." }		
28 Sep 15:10:08	STATUS_CHANGE	Kamran Ali	Claim CLM-FE40F6C0B8	{ "from": "Draft", "to": "Submitted", "note": "Claim submitted" }		
28 Sep 15:10:08	CLAIM_SUBMITTED	Hira Siddiqui Customer	Claim CLM-26CC70D39B	{ "first_submission": true }		
28 Sep 15:10:08	STATUS_CHANGE	System	Claim CLM-26CC70D39B	{ "from": "Under Evaluation", "to": "Manual Review", "note": "..."		
28 Sep 15:10:08	FINAL_DECISION	Hira Siddiqui Customer	Claim CLM-26CC70D39B	{ "decision": "Manual Review Required", "rule": "D04", "..."		
28 Sep 15:10:08	MODEL_PREDICTION	Hira Siddiqui Customer	Claim CLM-26CC70D39B	{ "python": { "predicted_class": "Manual Review", "top_confidence": "..."		
28 Sep 15:10:07	STATUS_CHANGE	System	Claim CLM-26CC70D39B	{ "from": "Submitted", "to": "Under Evaluation", "note": "Automated..." }		
28 Sep 15:10:07	STATUS_CHANGE	Hira Siddiqui	Claim CLM-26CC70D39B	{ "from": "Draft", "to": "Submitted", "note": "Claim submitted" }		
28 Sep 15:10:07	CLAIM_SUBMITTED	Sana Malik Service-center staff	Claim CLM-03538133C3	{ "first_submission": true }		
28 Sep 15:10:07	STATUS_CHANGE	System	Claim CLM-03538133C3	{ "from": "Under Evaluation", "to": "Rejected", "note": "Automated..." }		
28 Sep 15:10:07	FINAL_DECISION	Sana Malik Service-center staff	Claim CLM-03538133C3	{ "decision": "Likely Invalid", "rule": "D01", "policy_version": "..."		
		Sana Malik Service-center	Claim	{ "python": { "predicted_class": "Invalid		

17_audit_trail.jpg — Audit trail with filters and export (top of the page)



18_mobile_dashboard.jpg — Customer dashboard at 390 px



19_mobile_claim.jpg — Claim page at 390 px

Dashboard
Products
Claims

Models ready
99
UT
Customer

Claims / CLM-A3880D4BB4

ApexBook Pro 16

Motherboard failure - filed for Usman Tariq by Sana Malik

Approved
Low risk

Track progress
Download report (PDF)

CLM-A3880D4BB4
ApexBook Pro 16
Consumer Electronics

Decided in 1.00 s

Python model
v2.0.0

Valid Claim 0.89

Valid 0.89
Invalid 0.08
Manual Review 0.03

Teachable Machine
gtm-514702678b8c

Valid Claim 0.99

Valid 0.99
Invalid 0.01
Manual Review 0.00

0.50
0.25
0.75
min 0.60

0
1

0.10
Acceptable Match

Both predict Valid Claim; confidence difference 0.10.

Likely valid

Why this decision

Both models predict Valid and no blocking rule fired. Rule D07 matched (D01-D06 did not).

D01
D02
D03
D04
D05
D06
D07

► 16 warranty checks passed

Evidence still needed
None.

What moved the Python model
Each input replaced by its typical training value; bar = change in the probability of "Valid Claim".

Purchase price: 1899 (typical 619.22) +0.03
Days from fault to claim: 4 (typical 6) -0.02
Product category: Consumer Electronics (typical Industrial Tools) +0.01

What the image model saw

Claim summary

CLM-A3880D4BB4 - Consumer Electronics - Motherboard failure

Within coverage
not seen before

Reported in time
4 days - limit 30

Manufacturing Defect
covered cause - diag 0.86

Receipt
on file

Evidence
3 of 3 supporting

Serial matches
registered = evidence

Dates consistent
purchase = fault = claim

Invoice unique
not seen before

Repairs
0 previous

Purchased 2026-03-02 · claim 2026-09-28
Claim facts vs policy limits. Contains no model output or decision.

The card holds claim facts only — no prediction, confidence or decision. Teal outlines mark the tiles that most drove the image model's answer.

Invoice unique (not seen before) -0.98
Serial matches (registered = evidence) -0.98
Manufacturing Defect (covered cause · diag 0.86) -0.95

Pipeline timings (ms): preprocess 14 · python 823 · card 29 · gtm 128 · rules 1 · total 999

Evidence

Each file is fingerprinted (SHA-256); receipts are read by OCR

Receipt
Warranty card
Damage photo
Serial photo
Repair report missing

Warranty card
warranty_card.png · 10.3 KB · 1225b8f8e122 · 28 Sep 2026

Open

Extracted details — OCR not available for this file

Damage / fault photo
damage_photo.png · 10.7 KB · 579662361931 · 28 Sep 2026

Open

Claim details

Product
ApexBook Pro 16
PRD-4222001221

Category
Consumer Electronics

Serial
SN-APX-4471823

Purchased
02 Mar 2026 · City Electronics Mall

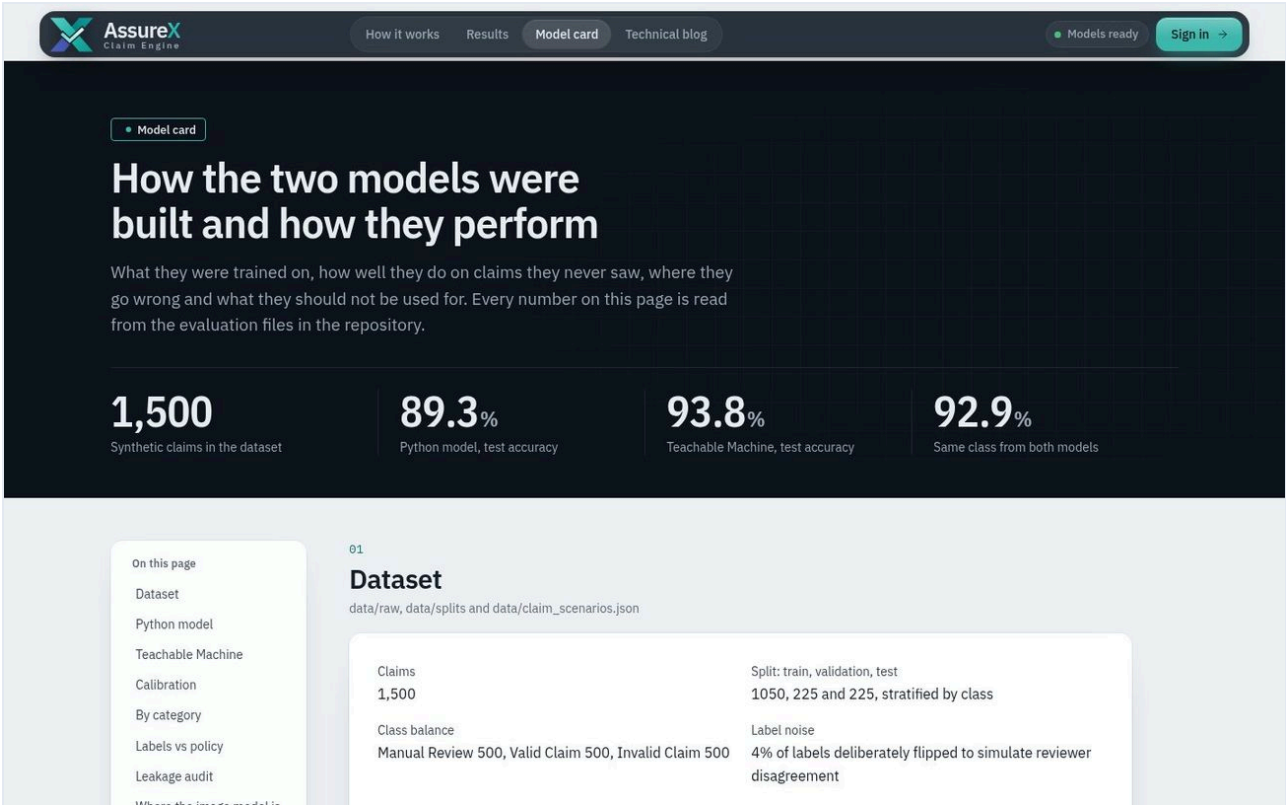
Warranty
Active until 02 Mar 2028

Fault started
24 Sep 2026

26_claim_likely_valid.jpg — Claim page, *Likely Valid* (rule D07): both models agree, no blocking rule fired (top of the page)

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27_model_card.jpg — Model card: both models' measured results as a technical report

Overview

Claims

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Audit

Models ready

AK Administrator

Overview > What-if

What-if simulator

Change when the two models count as agreeing and see the effect before it touches a single claim. Rule outcomes are replayed as stored; only the agreement check moves.

Decision policy 2026.09.26

On the 225 labelled test claims

Current thresholds

Measure	Current	Proposed	Change
Decided without a person	64.0%	64.0%	±0
Correct among those decisions	94.4%	94.4%	±0
Wrong automated decisions	8	8	±0
Sent to a reviewer	36.0%	36.0%	±0

Accuracy counts an automated "Likely Valid" as right when the true label is Valid Claim, and "Likely Invalid" when it is Invalid Claim. Claims sent to a reviewer are not counted as errors.

Where decisions move

0 of 225 test claims change

Rows: now · columns: with the proposed thresholds

	Valid	Invalid	Review
Valid	74	0	0
Invalid	0	70	0
Review	0	0	81

Live claims

0 of 11 evaluated claims would get a different recommendation

No live claim would get a different recommendation.

Thresholds

Minimum confidence 0.60

Below this, either model counts as unsure: Uncertain Result. Now 0.60.

Strong match: largest gap 0.10

Same class and confidences at most this far apart. Now 0.10.

Acceptable match: largest gap 0.25

Wider than this is a Weak Match and goes to a reviewer. Now 0.25.

Back to current

Apply thresholds

Applying saves a new decision-policy version with the change and its simulated effect in the audit trail.

Product

How a claim is decided

Measured results

Model card

Resources

Technical blog

Read it on Medium

GitHub repository

Python model v2.8.0

Teachable Machine gtm-514702678b8c

AssureX

AssureX Claim Engine · Built for TechWiz 7 · NextWave AI and ML

28_what_if_simulator.jpg — What-if simulator: try consistency thresholds against the stored decisions before applying them

Overview

Claims

Review queue

Analytics

Policies

Access

Audit

Models ready

AK Administrator

Overview > Batch

Batch evaluation

Run up to 500 claim records through both models, the warranty rules and the decision table. Nothing is saved as a claim; results are kept here for download.

Upload

CSV, UTF-8, at most 2 MB and 500 rows

Claim records

Same columns as the training splits. A claim_class column adds accuracy.

Choose File No file chosen

No file to hand? [Download the 225-claim test split](#) and upload it.

► [Required columns \(29\)](#)

Run the batch

Recent batches

No batches yet

Upload a file to run the first one.

Every warranty claim checked twice: a Python model, a Teachable Machine image model and a policy engine agree, or a person decides.

Python model v2.0.0

Teachable Machine gtm-514702678b8c

Product

How a claim is decided

Measured results

Model card

Resources

Technical blog

Read it on Medium

GitHub repository

AssureX

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29_batch_evaluation.jpg — Batch evaluation of uploaded claim records through the real pipeline

Dashboard
Products
Claims

Models ready
UT
Customer

Notifications

Claim updates, questions from reviewers and warranties about to end. Opening one marks it as read.

Unread 19
All 19

Unread

More information needed for CLM-5F9B6AD008

Reviewer request: Please upload a diagnostic report from an authorised center confirming the cause.

2m ago - CLM-5F9B6AD008

Review complete: CLM-03538133C3

Outcome: Rejected.

2m ago - CLM-03538133C3

Claim CLM-03538133C3 rejected

Automated evaluation rejected your claim. Reason: A policy hard-fail rule was triggered

2m ago - CLM-03538133C3

Claim CLM-03538133C3: Rejected

Recommendation: Likely Invalid. A policy hard-fail rule was triggered.

2m ago - CLM-03538133C3

Claim CLM-03538133C3 received

Your claim for AeroBreeze 1.5T was submitted and evaluated.

2m ago - CLM-03538133C3

Review complete: CLM-19E10D161E

Outcome: Rejected.

2m ago - CLM-19E10D161E

Claim CLM-19E10D161E rejected

Automated evaluation rejected your claim. Reason: A policy hard-fail rule was triggered

2m ago - CLM-19E10D161E

Claim CLM-19E10D161E: Rejected

Recommendation: Likely Invalid. A policy hard-fail rule was triggered.

2m ago - CLM-19E10D161E

Claim CLM-19E10D161E received

Your claim for NovaPhone 12 was submitted and evaluated.

2m ago - CLM-19E10D161E

Claim CLM-5F9B6AD008: Manual Review

Recommendation: Manual Review Required. A policy manual-review rule was triggered.

2m ago - CLM-5F9B6AD008

Claim CLM-5F9B6AD008 received

Your claim for VoltEdge Saw 18V was submitted and evaluated.

2m ago - CLM-5F9B6AD008

Review complete: CLM-E77387B57E

Outcome: Rejected.

2m ago - CLM-E77387B57E

Claim CLM-E77387B57E rejected

Automated evaluation rejected your claim. Reason: A policy hard-fail rule was triggered

2m ago - CLM-E77387B57E

Claim CLM-E77387B57E: Rejected

Recommendation: Likely Invalid. A policy hard-fail rule was triggered.

2m ago - CLM-E77387B57E

Claim CLM-E77387B57E received

Your claim for FrostGuard 450L was submitted and evaluated.

2m ago - CLM-E77387B57E

Review complete: CLM-A38B0D4BB4

Outcome: Approved.

2m ago - CLM-A38B0D4BB4

Claim CLM-A38B0D4BB4 approved

Automated evaluation approved your claim for ApexBook Pro 16.

2m ago - CLM-A38B0D4BB4

Claim CLM-A38B0D4BB4: Approved

Recommendation: Likely Valid. Both models predict Valid and no blocking rule fired.

2m ago - CLM-A38B0D4BB4

Claim CLM-A38B0D4BB4 received

Your claim for ApexBook Pro 16 was submitted and evaluated.

2m ago - CLM-A38B0D4BB4

Every warranty claim checked twice: a Python model, a Teachable Machine image model and a policy engine agree, or a person decides.

Python model v2.0.0

Teachable Machine gte-51270267886c

Product

How a claim is decided

Measured results

Model card

Resources

Technical blog

Read it on Medium

GitHub repository

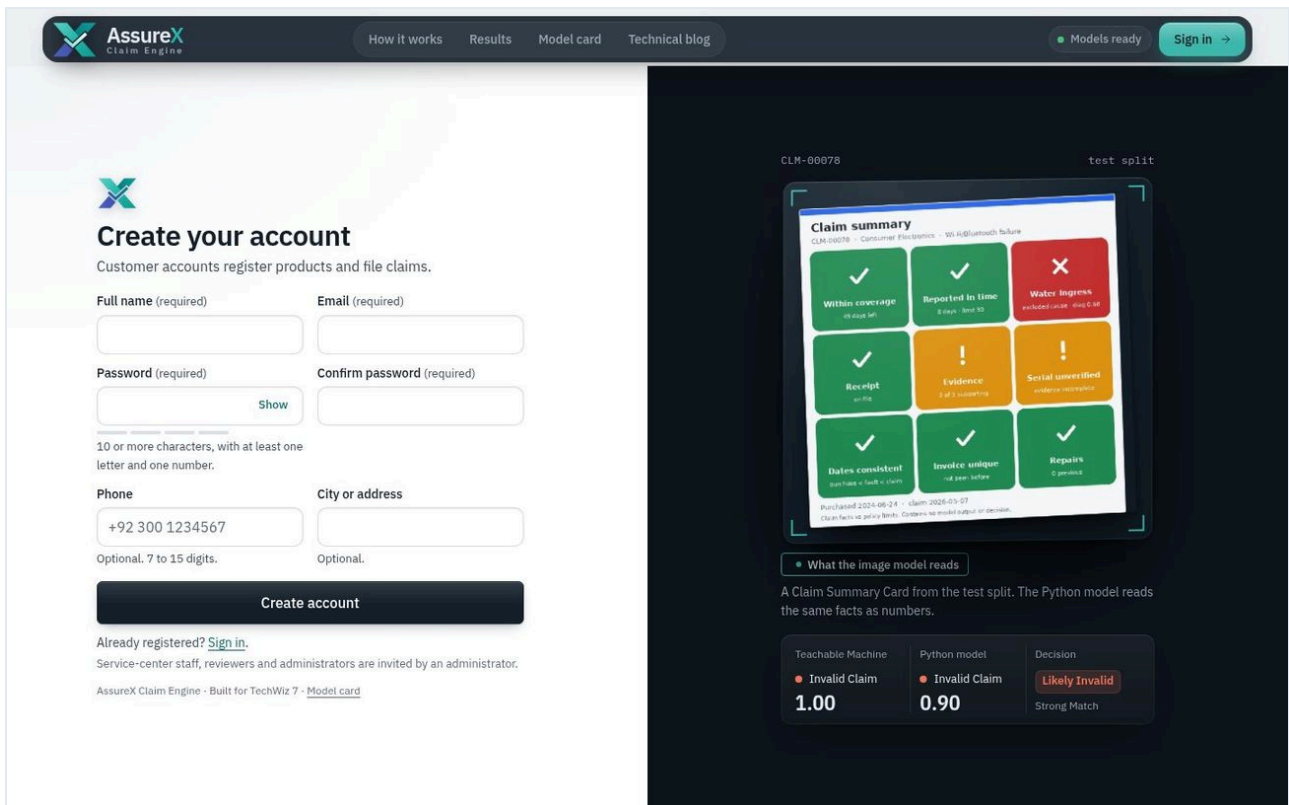
AssureX

AssureX Claim Engine - Built for TechWiz 7 - NextWave AI and ML

30_notifications.jpg — Notifications page with unread and all tabs

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31_register.jpg — Customer self-registration with field rules, password strength and a live match check

2. Deployed application

Item	Value
Public application URL	https://assurex.pythonanywhere.com
Source code	https://github.com/Saba1512006/assurex-claim-engine
Production server	<code>gunicorn wsgi:app</code> (Render: <code>render.yaml</code> ; PythonAnywhere: the WSGI file imports <code>application</code> from <code>wsgi.py</code>)
Health check	<code>/healthz</code> reports whether both models are loaded
Free-tier limitation	The free PythonAnywhere account runs Python 3.10 with a 512 MB disk quota, so both models can show as <i>Unavailable</i> there and claims then go to manual review; the full pipeline runs locally (see the installation instructions) and is shown in the demonstration video

Testing the deployed application

1. Open <https://assurex.pythonanywhere.com>. **Try a real claim** on the landing page runs three sample claims through both models and the rules live, without signing in. **Model card** shows how both models were measured.
2. **Sign in**: one click on an evaluator account on the sign-in page.
3. **Customer**: the dashboard lists what needs attention, product warranty plates and all claims; open a claim for the verdict (agreement gauge, both models, why, what the image model saw); *File a claim* runs the four-step wizard (a receipt PDF from any demo claim's evidence can be reused).
4. **Reviewer**: *Review queue > Start reviewing* opens the workbench. Keys: A approve, R reject, I request information, / comment, N next claim, ? help.
5. **Administrator**: *Overview* (KPIs and charts, each with a table view), *What-if simulator* (move the agreement thresholds and see the effect on 225 labelled claims before saving), *Batch evaluation* (upload the offered test-split CSV), *Policies* (change preview and version history), *Analytics*, *Models*, *Access*, *Audit*, CSV/Excel exports including every reviewer decision next to the models.
6. `/healthz` reports whether both models are loaded.

Evaluator accounts (demo data only)

Role	Email	Password	Can
Customer	customer@assurex.local	CustomerPass123!	Register products, file claims, upload evidence, track
Service-center staff	staff@assurex.local	StaffPass123!	File claims and register products for walk-in customers of <i>their</i> center, record repairs
Claim reviewer	reviewer@assurex.local	ReviewerPass123!	Work the manual-review queue, decide, request information, override with a reason
Administrator	admin@assurex.local	AdminPass123!	Dashboards, analytics, policies, thresholds, models, users, audit trail, exports

Public sign-up creates **customer** accounts only. Other roles are invited by an administrator (*Access* page) and must set their own password at first sign-in.